

Learning Assessment Header

The header depicted in the box below will appear at the top of every learning assessment. It includes formulas that do not need to be memorized and a summary of PsyBU math notation. Any formulas that you need, and that are not included in this header, are formulas you need to memorize.

Formulas:

$$\frac{sd}{\sqrt{n}} \quad \sqrt{\frac{sd_{pooled}^2}{n_1} + \frac{sd_{pooled}^2}{n_2}} \quad \sqrt{\frac{1-r^2}{n-2}} \quad \frac{sd_{est}}{sd_x \sqrt{n-1}} \quad \sqrt{\frac{t^2}{t^2+df}}$$
$$\bar{y} - b\bar{x} \quad b \frac{sd_x}{sd_y} \quad \frac{SS_{explained} - (df_{explained})(MS_{error})}{SS_{total} + MS_{error}}$$

Summary of PsyBU math notation:

Never type any spaces. Do not use formatting to change the font. Keep in mind standard order of operations: (a) parentheses, (b) exponents, (c) multiplication and division from left to right, (c) addition and subtraction from left to right. Use a caret symbol (^) to indicate superscripts and use an underscore (_) to indicate a subscripts.

$$x_i = x_i \quad x^2 = x^2 \quad \text{sqrt}(x) = \sqrt{x} \quad \text{sum}(x) = \Sigma(x)$$

$$(x)(y) = (x)(y) \quad x/y = \frac{x}{y} \quad x+y = x+y \quad x-y = x-y$$

$$\bar{x} = \bar{x} \quad \hat{x} = \hat{x} \quad \mu = \mu \quad \beta = \beta \quad \rho = \rho$$